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A PROJECT REPORT

ON

"To-Do List Web Application"

A project report submitted in partial fulfillment of the requirements for the degree of Bachelor of Technology in Computer Science and Engineering

BACHELOR OF TECHNOLOGY
COMPUTER SCIENCE AND ENGINEERING

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
AY 2025-26

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the Project work entitled "**To-Do List Web Application**" was carried out by the **Kashvi Balani, Atharva Jhibkate and Sanidhya Gholse** in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology** in **Computer Science and Engineering**, Symbiosis International (Deemed University), Pune, during the academic year 2025-2026.

Dr. Pavan Kumar Verma

DECLARATION

I hereby declare that the project titled "To-Do List Web Application" submitted to Symbiosis Institute of Technology, Constituent of Symbiosis International (Deemed University) Pune for the award of the degree of Bachelor of Technology in Computer Science and Engineering is a result of original research carried out by me. I understand that my report may be made electronically available to the public. It is further declared that the project report or any part thereof has not been previously submitted to any University or Institute for the award of any degree or diploma.

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ABSTRACT

This project presents the design and development of a feature-rich To-Do List Web Application built entirely using front-end technologies — HTML5, CSS3, and JavaScript. The application serves as an interactive daily task planner that enables users to efficiently organize, track, and manage their day-to-day activities. The interface features a dark-themed modern UI with animated backgrounds, a progress tracking ring, category-based task filtering (Morning, Work, Health, Personal), real-time statistics display, and persistent task storage using localStorage. Key features include dynamic task addition and deletion, task completion toggling with visual feedback, category-colored task indicators, motivational quotes, and a responsive layout compatible with all screen sizes. The project demonstrates practical application of core web technologies and modern UI/UX design principles, making it a useful productivity tool for students and working professionals alike.

Keywords: To-Do List, Task Management, HTML5, CSS3, LocalStorage, Productivity App, Web Application, Dark Theme UI, Daily Planner

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Chapter 1

Background and Technical Overview

1.1 Introduction

The To-Do List Web Application is a browser-based productivity tool designed to help users organize their daily tasks in a structured and visually engaging manner. Built using core web technologies — HTML5, CSS3, and vanilla JavaScript — the application requires no external frameworks or backend servers, making it lightweight, fast, and easily deployable. The project emerged from the recognition that many existing task management tools are overly complex, requiring account creation and internet connectivity even for basic use.

1.2 Technology Stack

- HTML5 — Structural markup with semantic elements, form inputs, SVG graphics
- CSS3 — Custom properties (CSS variables), animations (keyframes), Flexbox layout, responsive design
- Vanilla JavaScript (ES6+) — DOM manipulation, event listeners, localStorage API, dynamic rendering
- Google Fonts API — Baloo 2 and Quicksand typefaces for modern typography
- SVG — Circular progress ring rendered directly with inline SVG

1.3 System Requirements

The application runs entirely in the browser. It requires any modern web browser (Chrome, Firefox, Edge, Safari) with JavaScript enabled. No installation, server, or database setup is needed. localStorage support is required for task persistence between sessions.

Chapter 2

Problem Statement and Motivation

2.1 Problem Statement

Modern professionals and students face increasing challenges in managing their daily tasks effectively. Existing solutions such as Notion, Trello, or Google Tasks often require account registration, have steep learning curves, depend on internet connectivity, or carry unnecessary features that overwhelm simple use cases. There is a clear need for a lightweight, zero-dependency task manager that works offline, starts instantly, and presents tasks in a clear and motivating visual format.

2.2 Motivation

The motivation behind this project is to demonstrate that a fully functional, aesthetically pleasing, and practical productivity tool can be built using only fundamental web technologies. This project also serves as a demonstration of mastering HTML, CSS, and JavaScript without relying on frameworks like React or Vue, reflecting a deeper understanding of how the web platform works natively.

Chapter 3

Novelty and Innovative Contributions

The following features distinguish this project from generic To-Do list implementations:

3.1 Animated SVG Progress Ring

A dynamic circular SVG ring with gradient stroke visually represents the percentage of tasks completed in real time. The ring animates smoothly using stroke-dashoffset transitions, providing an intuitive and satisfying progress indicator absent in most basic to-do applications.

3.2 Category-Based Task Management

Tasks are classified into four life-area categories: Morning Routine, Work, Health, and Personal. Each category has a distinct color theme applied consistently across filter buttons, task left-border accents, and category tags, enabling users to visually scan tasks by domain at a glance.

3.3 Ambient Animated Background

Two floating gradient blobs animate continuously in the background using CSS keyframe animations with blur filters, creating a modern glassmorphism-inspired aesthetic that improves the visual experience without impacting performance.

3.4 Contextual Motivational Quotes

A rotating quote system randomly displays curated productivity quotes at the bottom of the page on each session load, providing psychological motivation to begin and complete tasks.

Chapter 4

Technical Advantages and Practical Usefulness

4.1 Technical Advantages

- Zero Dependencies: No frameworks, no build tools, no package managers required — just open the HTML file in a browser
- Offline Capable: Tasks persist via localStorage with no internet connection needed after initial load
- Performance: No JavaScript runtime overhead; executes directly in the browser engine
- Responsive Design: Fluid layout using CSS Flexbox and clamp() functions adapts to all screen sizes
- Single-File Deployment: The entire application lives in one .html file, trivially deployable to any web host or shared locally

4.2 Practical Usefulness

- Students can use it as a daily study planner without signing up for any service
- Professionals can bookmark it as a browser start page for instant task tracking
- Can be self-hosted on GitHub Pages, Netlify, or any static file server in under one minute
- Serves as a real-world teaching example for front-end web development courses

Chapter 5

Detailed Methodology / System Architecture

5.1 Architecture Overview

The application follows a single-page architecture (SPA) pattern without any routing library. All logic is encapsulated within a single HTML file. The data model is a JavaScript array of task objects stored in memory and mirrored to localStorage. The view is re-rendered on every state change by the render() function which rebuilds the DOM from scratch.

5.2 Data Model

Each task is represented as a JavaScript object with the following structure: { id: Number, text: String, cat: String (morning/work/health/personal), done: Boolean, time: String }. The array of tasks is serialized to JSON and stored in localStorage under the key 'vibrant_tasks'.

5.3 Core Functions

- render() — Filters tasks by active category, builds task card HTML, injects into DOM, calls updateStats()
- addTask() — Reads input field and category selector, creates a new task object, prepends to array, calls render()
- toggle(id) — Finds task by ID, flips done boolean, calls render()
- remove(id) — Filters out task by ID from array, calls render()
- updateStats() — Calculates total, done, pending counts and completion percentage; updates ring SVG, progress bar, and stats pills
- save() — Serializes tasks array to JSON and writes to localStorage

5.4 UI Component Structure

- Top Header — Greeting with animated wave emoji, live date string, and three stat pills (Total/Done/Left)
- Progress Strip — SVG ring with gradient, percentage label, linear progress bar, and contextual title
- Category Filter Row — Horizontally scrollable pill buttons with active state gradients
- Add Task Box — Text input, category dropdown, and animated + button
- Task List — Dynamically rendered task cards with color-coded left border, check button, category tag, time label, and delete button

Chapter 6

Prior Art and Related Work (Literature Survey)

Several existing tools and academic works in the task management domain were studied during the development of this project:

6.1 Commercial Tools

- Todoist — Cloud-based task manager with cross-device sync; requires account and internet, subscription for premium features
- Microsoft To Do — Integrated with Microsoft 365 suite; excellent for corporate users but heavyweight for standalone use
- Google Tasks — Minimalist task list tied to Google Calendar; lacks offline capability without the native app
- Notion — Feature-complete workspace tool; significant learning curve and overkill for simple daily planning

6.2 Academic References

- Garrett, J.J. (2010). The Elements of User Experience. New Riders — foundational UX principles applied in the UI design
- MDN Web Docs. (2024). Web Storage API — localStorage specification and usage patterns applied in this project
- W3C. (2023). CSS Animations and CSS Custom Properties specifications — applied in the animated background and theming system

6.3 Research Gap Addressed

The project addresses the gap between heavy commercial tools and overly basic tutorial-level implementations by delivering a production-quality UI with rich interaction patterns using only native browser APIs.

Chapter 7

Applications and Deployment Areas

7.1 Deployment Options

- Static Hosting: GitHub Pages, Netlify, Vercel — zero-cost deployment in under 5 minutes
- Local File: Simply opening the .html file in any browser provides full functionality
- Intranet: Can be hosted on a company's internal web server for team use without any server-side setup
- Offline PWA Extension: The codebase can be extended with a Service Worker manifest to function as a Progressive Web App

7.2 Target User Groups

- University students managing assignment deadlines and study schedules
- Freelancers tracking client deliverables without subscription costs
- Health and fitness enthusiasts tracking morning routines and workout goals
- Web development students learning DOM manipulation and localStorage through a practical project

Chapter 8

Conclusion and Future Scope

8.1 Conclusion

The To-Do List Web Application successfully demonstrates that sophisticated, interactive, and aesthetically refined productivity software can be built with zero external dependencies using only foundational web technologies. The application provides all core task management features — addition, completion tracking, deletion, categorization, and persistence — wrapped in a visually engaging dark-theme interface. The project reinforces the importance of JavaScript DOM manipulation, CSS animation, SVG graphics, and the browser's built-in storage APIs as powerful tools for real-world application development.

8.2 Future Scope

- Due Date & Reminders: Add date-time pickers per task with browser Notification API alerts
- Drag-and-Drop Reordering: Implement HTML5 Drag and Drop API or SortableJS for task prioritization
- PWA Conversion: Add service worker and web app manifest for installable offline-first app
- Dark/Light Mode Toggle: Implement a CSS variable-based theme switcher
- Backend Integration: Connect to a Node.js/Express REST API with MongoDB for multi-device sync
- AI Task Suggestions: Integrate a language model API to suggest task breakdowns or scheduling
- Export Functionality: Allow users to export their task list to PDF or CSV

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Appendices

Appendix A: Sample Task Data Structure (JSON)

```
{ "id": 1, "text": "Morning workout (30 min)", "cat": "health", "done":
false, "time": "6:00 AM" }
```

Appendix B: Feature Summary Table

Feature	Status
Task Addition & Deletion	✓ Implemented
Category Filtering	✓ Implemented
Progress Ring (SVG)	✓ Implemented
localStorage Persistence	✓ Implemented
Responsive Design	✓ Implemented
Motivational Quotes	✓ Implemented
Animated Background	✓ Implemented
Real-time Stats	✓ Implemented